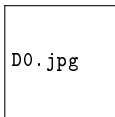


Lecture 37: ICA-5 Revision

Window Functions, Stored Procedures, Triggers

Z2004: Database Management Systems



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Plan for today

- ▶ MCQs: we'll justify the correct option and explain why others are wrong.
- ▶ SQL: we'll write a model solution, then list the most common exam mistakes.

Schema reminder: Student(SID, Name, Dept, GPA), Enroll(SID, CID, Semester, Marks), AuditLog(LogID, EventTime, EventType, TableName, KeyValue, Details).

Q1 (MCQ): PARTITION BY meaning

Correct: (b) restarts the rank separately for each department.

Why others are wrong

- ▶ (a) Filtering is WHERE, not PARTITION BY.
- ▶ (c) Sorting departments alphabetically is not what partition does.
- ▶ (d) Duplicate removal is DISTINCT, not partitioning.

Q2 (MCQ): DENSE_RANK vs RANK vs ROW_NUMBER

Correct: (c) DENSE_RANK().

Why

- ▶ Ties share the same rank.
- ▶ Next rank is consecutive (no gaps).

Why others are wrong

- ▶ ROW_NUMBER() never ties.
- ▶ RANK() creates gaps after ties.
- ▶ NTILE() assigns buckets, not ranks.

Q3 (MCQ): LAG with PARTITION + ORDER

Correct: (a) previous row within the same Dept ordering.

Why others are wrong

- ▶ (b) “next row” is LEAD, not LAG.
- ▶ (c) max needs MAX().
- ▶ (d) average needs AVG().

Q4 (MCQ): What is a trigger?

Correct: (c) code that automatically runs when an INSERT/UPDATE/DELETE occurs.

Why others are wrong

- ▶ (a) That describes a normal query (manual execution).
- ▶ (b) Stored query is closer to a view; triggers don't replace indexes.
- ▶ (d) Backup/recovery is a DB admin feature, not a trigger.

Q5: DENSE_RANK within each department (model SQL)

```
SELECT SID, Name, Dept, GPA,  
       DENSE_RANK() OVER (PARTITION BY Dept ORDER BY GPA DESC) AS DeptRank  
FROM Student;
```

Exam mistakes:

- ▶ Forgetting PARTITION BY Dept (gives a global rank).
- ▶ Ordering ASC (lowest GPA becomes rank 1).
- ▶ Using ROW_NUMBER when ties should share a rank.

Q6: LAG for PersonAhead + GapToAhead (model SQL)

```
SELECT SID, Name, Dept, GPA,  
       LAG(Name) OVER (PARTITION BY Dept ORDER BY GPA DESC) AS PersonAhead,  
       LAG(GPA) OVER (PARTITION BY Dept ORDER BY GPA DESC) - GPA AS GapToAhead  
FROM Student;
```

Exam mistakes:

- ▶ Missing PARTITION BY Dept (PersonAhead comes from another dept).
- ▶ Using LEAD instead of LAG.
- ▶ Computing the gap as $GPA - LAG(GPA)$ (sign flipped).

Q7: Stored procedure sp_TopNStudentsByDept (model)

One acceptable SQL Server-style answer (dialect variations OK):

```
CREATE PROCEDURE sp_TopNStudentsByDept
    @Dept VARCHAR(20),
    @N     INT
AS
BEGIN
    SELECT TOP (@N) SID, Name, GPA
    FROM Student
    WHERE Dept = @Dept
    ORDER BY GPA DESC;
END;
```

Exam mistakes:

- ▶ Not ordering by GPA DESC.
- ▶ Forgetting to filter by Dept.
- ▶ Hard-coding N instead of using the parameter.

Q8: AFTER INSERT trigger to write AuditLog (model)

One acceptable SQL Server-style answer (dialect variations OK):

```
CREATE TRIGGER trg_Enroll_Audit
ON Enroll
AFTER INSERT
AS
BEGIN
    INSERT INTO AuditLog(EventTime, EventType, TableName, KeyValue, Details)
    SELECT CURRENT_TIMESTAMP,
           'INSERT',
           'Enroll',
           CAST(i.SID AS VARCHAR(50)),
           CONCAT('CID=', i.CID, ', Semester=', i.Semester)
    FROM inserted i;
END;
```

Exam mistakes:

- ▶ Writing a trigger that only works for single-row inserts.
- ▶ Forgetting to insert EventType/TableName.
- ▶ Not including both CID and Semester in Details.

- ▶ Window functions: always ask **partition?** **order?** **ties?**.
- ▶ Procedures: parameterize + filter + order + limit.
- ▶ Triggers: think in **sets** (multi-row DML) and log required fields.

Overall performance summary (Assignments 1–5, Quizzes 1–2)

Assignments (marks):

Component	Min	Max	Avg
Assignment 1	8	19.5	14.6
Assignment 2	0	19.5	10.5
Assignment 3	14	20	19.5
Assignment 4	2	15	10.1
Assignment 5	0	19	11.8

Quizzes (final, /15):

Component	Min	Max	Avg	Std
Quiz 1 (Final /15)	0	14.7	11.3	2.99
Quiz 2 (Final /15)	1.8	13.8	8.1	3.20

Attendance bonus: Students with **100% attendance** will receive **attendance bonus marks**.

Grade weights + current total (computed so far)

Course weightage:

- ▶ Assignments (A1–A5): **10%**
- ▶ Quizzes (Q1–Q2): **30%**
- ▶ Project: **20%** (not yet included)
- ▶ End-sem exam: **40%** (not yet included)

Current total computed so far (out of 40):

- ▶ Assignment contribution (/10):

$$\frac{A1 + A2 + A3 + A4 + A5}{100} \times 10 = \frac{A1 + A2 + A3 + A4 + A5}{10}$$

- ▶ Quiz contribution (/30):

$$Q1_{\text{final}} + Q2_{\text{final}}$$

- ▶ **Current total (/40):**

$$\frac{A1 + \dots + A5}{10} + (Q1_{\text{final}} + Q2_{\text{final}})$$

Current total (/40): top 5 and bottom 5 (no IDs/names)

Top 5 (current total /40):

- 1 36.75
- 2 35.15
- 3 34.90
- 4 32.70
- 5 32.40

Bottom 5 (current total /40):

- 1 3.50
- 2 11.25
- 3 14.65
- 4 16.00
- 5 19.70

Next week: project completion + report cards

- ▶ Next week (after project evaluation), your **overall total will be computed including the Project**.
- ▶ You will receive an individual **report card by email**.
- ▶ The report card will show:
 - Your marks **out of 60** (Assignments 10 + Quizzes 30 + Project 20),
 - **Attendance status**,
 - **Attendance bonus: +3 marks** for students with **100% attendance**.

End-sem exam: format + lab exam rules

Exam components (format not finalized yet):

- ▶ **SQL query-writing practical** using **DBeaver** .
- ▶ **Theory section** will also be included.
- ▶ Detailed instructions for the lab exam will be shared separately.

SQL practical instructions (important):

- ▶ Type **all queries** in a single **.sql** file.
- ▶ Rename the file to **your roll number** before submission.
- ▶ **Internet will be switched off.**
- ▶ **No talking. No copying. No phones.**

Academic integrity:

- ▶ Cheating in the SQL practical $\Rightarrow$ **0 marks for the practical component.**
- ▶ Cheating in theory $\Rightarrow$ **reported to the Academic Committee** with recommendation for strict action (which may include **failing the exam/course**).

Project presentations + exam instructions: timeline

- ▶ **Project presentation instructions and schedule** will be shared on **Google Classroom** before **EOD today**.
- ▶ **All end-sem exam instructions** will be shared by **Wednesday EOD next week** (the **last day of instruction**).

Questions?